



CPA Exam analytics definitions and valuation metrics

Many ratios and metrics used by accountants can be calculated in more than one way. The Analytics Definitions and Valuation Metrics below provide the formulas used for many of the ratios and metrics tested in Core and Discipline sections of the Uniform CPA Examination. This resource provides the formulas used in the questions unless a question explicitly states otherwise.

These formulas are not provided with the multiple-choice questions (MCQs) in the first two testlets but are eligible for testing to the extent they are relevant to an Area, Group, Topic, or representative task in the Blueprints. The applicable formulas are provided in specific task-based simulations (TBSs) and candidates should use the formulas given in the TBS to complete the task.

Analytics definitions

Analytic	Formula
Accounts receivable turnover	$\text{Sales (net)} \div \text{Average accounts receivable (net)}$
Asset turnover	$\text{Sales (net)} \div \text{Average total assets}$
Basic earnings per share	$\text{Income available to common shareholders} \div \text{Weighted-average common shares outstanding}$
Cash conversion cycle	$\text{Days sales in accounts receivable} + \text{Days in inventory} - \text{Days of payables outstanding}$
Current ratio	$\text{Current assets} \div \text{Current liabilities}$
Days in inventory	$\text{Ending inventory} \div (\text{Cost of goods sold} \div 365)$
Days of payable outstanding	$\text{Ending accounts payable} \div (\text{Cost of goods sold} \div 365)$
Days sales in accounts receivable	$\text{Ending accounts receivable (net)} \div (\text{Sales (net)} \div 365)$
Debt to equity	$\text{Total liabilities} \div \text{Total equity}$
Dividend payout	$\text{Cash dividends} \div \text{Net income}$
Equity multiplier	$\text{Total assets} \div \text{Total equity}$
Gross margin (Gross profit margin)	$(\text{Sales (net)} - \text{Cost of goods sold}) \div \text{Sales (net)}$
Inventory turnover	$\text{Cost of goods sold} \div \text{Average inventory}$
Operating cash flow ratio	$\text{Cash flow from operations} \div \text{Ending current liabilities}$
Price earnings ratio	$\text{Price per share} \div \text{Basic earnings per share}$
Profit margin	$\text{Net income} \div \text{Sales (net)}$
Quick ratio	$(\text{Cash and cash equivalents} + \text{Short-term marketable securities} + \text{Receivables (net)}) \div \text{Current liabilities}$
Return on assets	$\text{Net income} \div \text{Average total assets}$
Return on equity	$\text{Net income} \div \text{Average total equity}$
Return on sales	$\text{Income before interest income, interest expense, and taxes} \div \text{Sales (net)}$
Times interest earned	$\text{Income before interest expense and taxes} \div \text{Interest expense}$ or $\text{Earnings before interest and taxes} \div \text{Interest expense}$
Total debt ratio	$\text{Total liabilities} \div \text{Total assets}$

Valuation metrics

Metric	Formula
Capital Asset Pricing Model (CAPM)	$E_r = R_f + B \times (R_m - R_f)$ where: E_r = Expected return on an asset or portfolio of assets R_f = Risk-free rate of return R_m = Expected return on a market portfolio B = Beta
Discounted cash flow valuation	$PV = CF_1 \div (1 + r)^1 + CF_2 \div (1 + r)^2 + CF_3 \div (1 + r)^3 + \dots + CF_t \div (1 + r)^t$ where: PV = Present value CF = Cash flow r = Discount rate t = Final period
Dividend discount model	$P_0 = D_1 \div (1 + r)^1 + D_2 \div (1 + r)^2 + D_3 \div (1 + r)^3 + \dots + D_t \div (1 + r)^t$ where: P_0 = Stock price D = Dividend r = Discount rate t = Final period
Dividend growth model	$P_0 = [D_0 \times (1 + g)] \div (r - g)$ where: P_0 = Current price of a stock D_0 = Current dividend r = Discount rate g = Dividend growth rate
Economic Value Added (EVA)	After-tax operating income – [Weighted-average cost of capital × (Total assets – Current liabilities)]
Gross Domestic Product (GDP) - Nominal	$C + I + G + X$ where: C = Personal consumption expenditures I = Gross private domestic investment G = Government purchases X = Net exports
GDP - Real	Nominal GDP ÷ GDP Deflator
Internal growth rate	$(\text{Return on assets} \times \text{Retention}) \div (1 - \text{Return on assets} \times \text{Retention})$ where: $\text{Retention} = \text{Addition to retained earnings} \div \text{Net income}$
Internal rate of return	Discount rate that makes the Net Present Value (NPV) of an investment zero
Net Present Value (NPV)	Discounted cash flow valuation (see formula above) – Initial investment

Valuation metrics (continued)

Metric	Formula
Payback period	Initial investment ÷ Annual incremental cash flow where: The cash flow per period is even
Profitability index	Present value of cash flows ÷ Cost of investment
Residual income	Income – (Required rate of return × Investment)
Value of a levered firm	Value of an unlevered firm + Present Value (PV) of the interest tax savings where: Interest tax savings = $T_c \times (r_{\text{debt}} \times D)$ PV of the interest tax savings = $T_c \times (r_{\text{debt}} \times D) \div r_{\text{debt}}$ assuming the debt is permanent T_c = Corporate tax rate r_{debt} = Rate of interest on the debt D = Amount of debt
Weighted-average cost of capital	$(E \div V) \times R_e + (P \div V) \times R_p + (D \div V) \times R_d \times (1 - T)$ where: E = Market value of common stock equity V = E + P + D R_e = Required return on common stock equity P = Market value of preferred stock equity R_p = Required return on preferred stock equity D = Market value of debt R_d = Required return on debt T = Corporate tax rate

